

Navigating the Dilemma: Sterilization and Monetary Transmission in Emerging Markets

Priyansh Mahendra¹

¹Ahmedabad University, priyansh.m2@ahduni.edu.in

Abstract

Rey’s (2013) “Dilemma” hypothesis argues that free capital mobility alone is enough to erode a country’s monetary policy independence, regardless of its exchange-rate regime – a particularly acute concern for emerging market economies (EMEs) exposed to volatile foreign capital flows. This paper tests whether foreign capital inflows weaken monetary policy transmission in India, and whether the Reserve Bank of India’s (RBI) liquidity sterilization operations offer a way to counteract this effect. Using quarterly data from 2014Q1–2025Q3, we construct a novel Money Source Index (MSI) to capture foreign-sourced liquidity growth and a time-varying monetary transmission score derived from a rolling Time-Varying Parameter VAR (TVP-VAR). Contrary to a simplistic reading of Rey’s Dilemma, foreign capital inflows are associated with *stronger*, not weaker, transmission ($\beta_1 = 0.72$, $p < 0.01$). A Baron & Kenny mediation analysis shows this effect operates almost entirely through RBI’s sterilization response: once sterilization intensity is included in the model, the direct effect of capital inflows becomes statistically insignificant ($\delta_1 = 0.04$, $p = 0.75$), while sterilization remains a strong, significant predictor ($\delta_2 = -0.02$, $p < 0.01$). These findings suggest that active balance-sheet management offers EMEs a practical “third way” to preserve monetary autonomy amid large capital flows, without resorting to capital controls.

Keywords: Impossible Trinity, Global Financial Cycle, Monetary Transmission, Sterilization

1 Introduction

The Mundell–Fleming “Trilemma” holds that a country can achieve at most two of three policy goals at once: an independent monetary policy, a fixed exchange rate, and free cross-border capital flows. Rey (2013) challenged this view, arguing that a global financial cycle – driven largely by U.S. monetary policy – transmits to nearly every country regardless of its exchange-rate regime, so that free capital mobility on its own is enough to erode monetary independence. This collapses the three-way trade-off into a “Dilemma”: independent monetary policy is possible only if the capital account is actively managed. For emerging market economies (EMEs), which experience sharper surges and reversals of foreign capital than advanced economies, this raises a pressing question: when foreign capital floods in, does the central bank lose control of domestic monetary policy? And if so, what can it do to get that control back? This paper uses a Time-Varying Parameter VAR (TVP-VAR) and a mediation framework on Indian data to show that active liquidity sterilization by the central bank is one concrete way to navigate Rey’s dilemma.

2 Literature Review

2.1 The Global Financial Cycle and Rey’s Dilemma

Rey (2013) documented the existence of a global financial cycle in asset prices, credit growth, and capital flows that is closely tied to U.S. Federal Reserve policy and to global risk appetite. Under this view, a country’s domestic monetary conditions can be destabilized by foreign capital waves largely irrespective of whether its currency floats or is pegged. The taper-tantrum episode of 2013, which hit several EMEs including Brazil, Turkey, and Indonesia, is widely cited as an example consistent with this dynamic.

2.2 Monetary Policy Transmission in EMEs

Textbook monetary transmission implies that when a central bank raises or cuts its policy rate, commercial banks pass this through to their own lending and deposit rates in reasonably close proportion. In practice, several frictions can weaken this pass-through in India: banks are slow to cut lending rates when they are still carrying high-cost term deposits raised earlier (elevated cost of funds); banks with large stocks of non-performing assets keep lending rates artificially high to protect interest income; and administered-rate government savings schemes create a floor below which banks are reluctant to cut deposit rates, since doing so would trigger deposit outflows into these schemes.

2.3 Gap and Hypotheses

A specific transmission problem arises when a large wave of foreign capital enters India. Foreign investors sell dollars and buy rupees to invest in Indian assets. Left unaddressed, the resulting demand for rupees would cause rapid appreciation, hurting Indian exporters. To prevent this, the Reserve Bank of India (RBI) buys the incoming dollars and pays out newly created rupees, injecting fresh liquidity into the banking system. If left in the system, this liquidity surplus risks fuelling inflation and – because banks now have ample cheap funding of their own – weakens their need to follow RBI’s policy-rate signals, undermining monetary transmission. To manage this, the RBI can sterilize the inflow: it sells government securities to banks, draining back the rupees it created and neutralizing the liquidity impact. This paper tests two hypotheses that speak directly to this mechanism:

1. H1: Foreign capital inflows impair domestic monetary transmission.
2. H2: Foreign capital inflows trigger active sterilization by the RBI, which in turn strengthens transmission.

3 Objectives

This paper’s primary objective is to examine how foreign capital inflows affect the strength of monetary policy transmission in India, and to test whether RBI’s liquidity sterilization operations are the specific channel through which any such effect operates.

4 Methodology

4.1 Sample Period and Data Frequency

This is an empirical study using quarterly time-series data. The underlying source series were compiled from RBI’s Database on the Indian Economy (DBIE), NSDL’s FPI reports, the ACE Knowledge Portal, and Yahoo Finance, and were then cleaned and merged into a model-ready panel of **45 complete quarters, running from 2014Q1 to 2025Q3**. Earlier quarters (pre-2014) were dropped because several core series – most notably CPI inflation on the new base year and the current account balance – had extensive missing data before that point; the current account variable was dropped altogether due to persistent post-2019 gaps. Key variables include:

1. Quarterly change in the benchmark policy repo rate set by RBI
2. Quarterly change in Consumer Price Index (CPI) inflation
3. Quarterly growth in broad money supply (M3), measuring total liquidity expansion in the system
4. Quarterly change in commercial banks’ prime lending rate (PLR midpoint), the transmission variable of interest
5. The Money Source Index (MSI): the ratio of the trailing four-quarter sum of foreign capital inflows (FPI debt plus NRI deposits) to the trailing four-quarter sum of broad money expansion, capturing how much of India’s liquidity growth is foreign-sourced
6. Net Liquidity Adjustment Facility (LAF) operations, measuring the scale of RBI’s liquidity injection (positive) or absorption (negative)

7. Year-on-year change in the quarterly average Brent crude oil price, a control for imported-inflation and terms-of-trade shocks
8. GovShare: the share of broad money growth over the trailing four quarters that was absorbed by central and state government market borrowing, a control for fiscal crowding-out effects on liquidity

4.2 Research Design

A Time-Varying Parameter VAR (TVP-VAR) is used to capture how the relationship between policy rates and lending rates evolves over time, since a fixed-coefficient model would blend together very different monetary regimes – inflation targeting, COVID-era easing, and the post-2019 external benchmark lending rate (EBLR) regime – into a single, potentially misleading, average effect. A mediation model then tests whether RBI’s sterilization operations are the specific mechanism linking foreign inflows to transmission strength. The analysis proceeds in six phases: (1) data preparation; (2) stationarity testing of the four core VAR variables (repo rate, CPI inflation, M3 growth, and PLR), with non-stationary series first-differenced; (3) selecting the Cholesky ordering and lag length for the VAR; (4) estimating a time-varying transmission score using a 24-quarter rolling window with 1 lag (as recommended by the AIC criterion); (5) testing H1 – whether foreign inflows (MSI) affect the transmission score; and (6) testing H2 using the Baron & Kenny three-step mediation framework to establish whether RBI’s liquidity sterilization mediates the relationship between foreign inflows and transmission.

4.3 Analytical Techniques

4.3.1 Transmission Score

The transmission score is the four-quarter cumulative response of commercial bank lending rates (PLR) to a 100-basis-point policy-rate shock, generated from an orthogonalized impulse response function (IRF) within a rolling VAR(1) estimated over a 24-quarter (6-year) window ending in each quarter. For example, a score of 0.66 in 2020Q2 means a 100-bps repo hike would be associated with only a 66-bps rise in lending rates over the following year – i.e., weak transmission. Because the four core VAR variables (repo rate, CPI inflation, M3 growth, PLR) were non-stationary in levels (ADF test), each was replaced with its own quarterly change before estimation, and CPI inflation, M3 growth, and PLR were confirmed stationary in changes at conventional significance levels.

4.3.2 Mediation Analysis

The Baron & Kenny three-step framework is used to test the proposed sterilization channel. It isolates the indirect effect of capital inflows that runs through RBI’s liquidity absorption from any direct effect of capital inflows on transmission, allowing the two to be statistically distinguished.

5 Results

5.1 Baseline Findings

The contemporaneous impact of foreign capital inflows (MSI) on the transmission score is significantly positive ($\beta_1 = 0.8231$, $p < 0.01$). Because capital inflows and transmission strength could plausibly move together for reasons unrelated to sterilization (simultaneity), the model is re-estimated using the one-quarter lagged value MSI_{t-1} with Newey–West HAC standard errors to correct for serial correlation. The lagged effect remains significantly positive ($\beta_1 = 0.7150$, $p < 0.01$), and a Granger causality test confirms the estimate is not an artifact of reverse causation from transmission strength to MSI (though, as discussed below, the same test does not find that MSI Granger-causes the transmission score at conventional lags). *This positive sign runs counter to a simplistic reading of Rey’s Dilemma, in which more foreign inflow should erode, not strengthen, transmission.*

5.1.1 Contemporaneous Baseline Model

$$\text{Score}_t = \beta_0 + \beta_1 MSI_t + \beta_2 \text{GovShare}_t + \beta_3 \text{Brent}_t + \beta_4 \text{EBLR}_t + \varepsilon_t \quad (1)$$

5.1.2 Lagged Specification (Endogeneity Control)

To address potential simultaneity and reverse-causality concerns, the model replaces contemporaneous MSI_t with its one-quarter lagged value MSI_{t-1} :

$$Score_t = \beta_0 + \beta_1 MSI_{t-1} + \beta_2 GovShare_t + \beta_3 Brent_t + \beta_4 EBLR_t + \varepsilon_t \quad (2)$$

Results remain consistent with the baseline: the coefficient stays positive and significant at the 1% level.

5.2 Mediation Analysis Results

The three links below test whether capital inflows operate on transmission specifically through the sterilization channel. Figure 1 traces the full chain over time: foreign inflow pressure (Panel A) is followed by a corresponding rise in RBI's sterilization intensity (Panel B), while the transmission score (Panel C) stays close to or above full pass-through even as inflows persist.

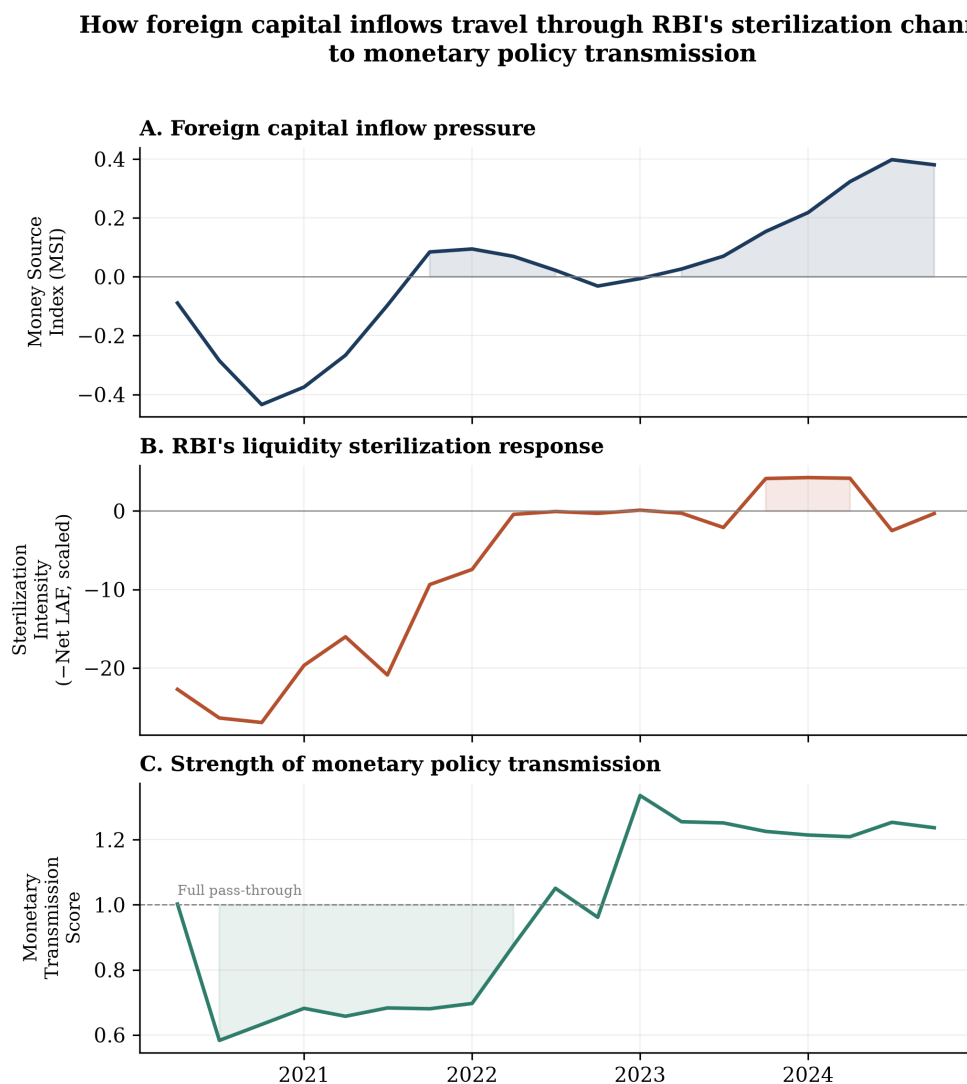


Figure 1: The sterilization channel over time. Panel A shows the Money Source Index (MSI), the paper's measure of foreign capital inflow pressure. Panel B shows RBI's sterilization intensity ($-Net\ LAF$; higher values indicate more liquidity absorption). Panel C shows the resulting monetary transmission score, with the dashed line marking full pass-through (score = 1).

5.2.1 Link 1: Capital Inflow to Liquidity Absorption (MSI → LAF)

Higher lagged foreign inflows are associated with significantly lower Net LAF ($\alpha_1 = -33.28, p < 0.01$). *Note: a negative coefficient here does not indicate a reversal in direction – a lower (more negative) Net LAF value means RBI is absorbing more liquidity, so the negative sign reflects an accounting convention, not a weaker relationship.*

$$\text{LAF}_t = \alpha_0 + \alpha_1 \text{MSI}_{t-1} + \alpha_2 \text{GovShare}_t + \alpha_3 \text{Brent}_t + \alpha_4 \text{EBLR}_t + u_{1,t} \quad (3)$$

This is consistent with the mechanism described in Section 3.3: more foreign inflow forces RBI into more active sterilization.

5.2.2 Link 2: Liquidity Absorption to Transmission Efficiency (LAF → Score)

Greater liquidity absorption (i.e., a lower Net LAF) is associated with a significantly stronger transmission score ($\gamma_1 = -0.0195, p < 0.01$).

$$\text{Score}_t = \gamma_0 + \gamma_1 \text{LAF}_t + \gamma_2 \text{GovShare}_t + \gamma_3 \text{Brent}_t + \gamma_4 \text{EBLR}_t + u_{2,t} \quad (4)$$

5.2.3 Link 3: Horse-Race / Full Mediation Model

When MSI_{t-1} and Net LAF are entered together, the direct effect of MSI becomes small and statistically insignificant ($\delta_1 = 0.0434, p = 0.752$), while Net LAF remains strong and significant ($\delta_2 = -0.0195, p < 0.01$).

$$\text{Score}_t = \delta_0 + \delta_1 \text{MSI}_{t-1} + \delta_2 \text{LAF}_t + \delta_3 \text{GovShare}_t + \delta_4 \text{Brent}_t + \delta_5 \text{EBLR}_t + u_{3,t} \quad (5)$$

This pattern – a significant total effect of MSI alone, but an insignificant direct effect once LAF is included, alongside a significant LAF coefficient – is the signature of full mediation: capital inflows appear to affect transmission almost entirely *through* RBI’s sterilization response, not directly. Figure 2 makes this contrast visible: once both variables compete in the same regression, the slope on capital inflow flattens out while the slope on sterilization remains steep and significant.

The horse race: once both effects compete, sterilization wins

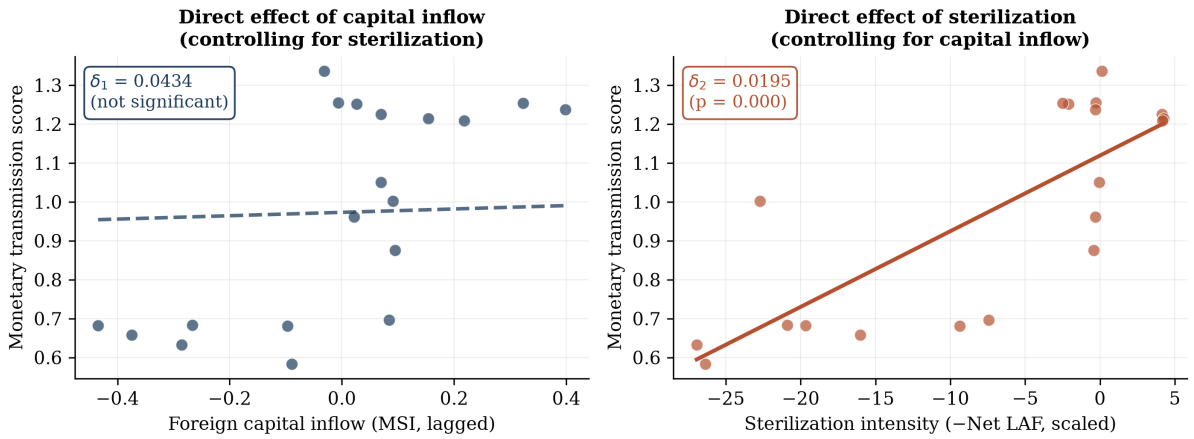


Figure 2: The horse race between capital inflow and sterilization (Section 5.2, Link 3). Each panel plots the transmission score against one variable, with the slope drawn from the horse-race coefficient (δ_1, δ_2) rather than a raw univariate fit. The left panel shows the weak, statistically insignificant direct effect of foreign capital inflow once sterilization is controlled for; the right panel shows the strong, significant effect of sterilization intensity that survives the same test.

5.3 Robustness Checks

Three checks support the main result. First, controlling for the post-2019 EBLR lending-rate regime (which mechanically raised pass-through), the MSI effect persists and remains significant ($\beta_1 = 0.8297$, $p < 0.01$). Second, using lagged rather than contemporaneous MSI, as reported above, addresses simultaneity concerns and leaves the result intact. Third, re-estimating the rolling VAR and lagged-MSI regression across alternative window lengths of 4, 5, and 6 years yields a significant, positive MSI coefficient in 2 of the 3 windows (5-year and 6-year), while the shortest 4-year window is not statistically significant – plausibly reflecting the smaller number of usable rolling-window observations (27 vs. 19–23) once a shorter window is used. Notably, a standard Granger causality test does not find that MSI Granger-causes the transmission score at 1 or 2 lags ($p = 0.25$ and $p = 0.28$), so the regression-based evidence for H1 and H2 should be read as evidence of a robust statistical association rather than definitive proof of predictive causal ordering; this is flagged as a limitation below.

6 Discussion

The results suggest RBI is not a passive victim of Rey’s global financial cycle. Instead, foreign capital inflows appear to prompt an active policy response – sterilization – that, on this evidence, is associated with restored and even strengthened monetary transmission rather than a weakened one. Active liquidity management may therefore offer EMEs like India a practical “third way” to manage Rey’s dilemma without resorting to blunt capital controls. That said, the evidence is associational: the sample is short (45 quarters), the mediation tests rely on the same estimated transmission score used throughout, and the Granger test leaves open questions about the precise timing of causality. Future work with a longer sample, alternative transmission measures, and instrumental-variable identification would help confirm the causal channel proposed here.

7 Conclusion

Rey’s dilemma is real, but it is not a binary choice between full capital controls and lost monetary autonomy. This paper’s evidence indicates that active balance-sheet management – specifically, liquidity sterilization – gives emerging market central banks a practical tool to preserve, and in India’s case apparently strengthen, monetary policy transmission even amid large foreign capital inflows. For policymakers, this points to sterilization capacity and operational sophistication in liquidity management as first-order tools for macroeconomic stability in a world dominated by the global financial cycle.

References

1. Rey, H. (2013). Dilemma not trilemma: The global financial cycle and monetary policy independence. *Proceedings of the Federal Reserve Bank of Kansas City Economic Policy Symposium*, 285–333.
2. Patra, M. D., Bhattacharyya, I., John, J., & Kumar, A. (2024). Monetary policy transmission in India: The recent experience. *Reserve Bank of India Bulletin*, 78(10), 189–208.
3. Mohan, R., & Patra, M. D. (2009). Monetary policy transmission in India. In *BIS Papers: Vol. 35. Monetary policy and the measurement of inflation: Prices, wages and expectations* (pp. 259–309). Bank for International Settlements. <https://www.bis.org/publ/bppdf/bispap35m.pdf>
4. Primiceri, G. E. (2005). Time varying structural vector autoregressions and monetary policy. *The Review of Economic Studies*, 72(3), 821–852. <https://doi.org/10.1111/j.1467-937X.2005.00353.x>
5. Sengupta, R. (2014). *Monetary policy transmission in India: A vector autoregression analysis* (IMF Working Paper No. WP/14/156). International Monetary Fund. <https://www.imf.org/external/pubs/ft/wp/2014/wp14156.pdf>
6. Kumawat, L. (2024). Time-variation in response of inflation to monetary policy shocks in India: Evidence from TVP-VAR models. *Indian Economic Review*, 59(1), 145–174. <https://doi.org/10.1007/s41775-024-00218-y>

7. Phul, S. (2024). Efficacy of monetary policy transmission during the flexible inflation targeting regime in India. *South Asian Journal of Macroeconomics and Public Finance*, 13(1), 70–99. <https://doi.org/10.1177/22779787241245237>
8. Raghuvanshi, A. P., & Ahmad, W. (2024). Inflation targeting and monetary policy response in India. *Asian Economics Letters*. <https://doi.org/10.1016/j.asieco.2024.101822>
9. Alex, D. (2025). Inflation targeting and the changing transmission mechanism of monetary policy in India. *Economic Modelling*, 151, 107141. <https://doi.org/10.1016/j.econmod.2025.107141>
10. Kumawat, L. (2025). Transmission of monetary policy shocks to inflation in India: An empirical investigation of different channels. *Journal of Quantitative Economics*. <https://doi.org/10.1177/09767479251360213>
11. Joseph, J. P., & Dash, S. K. (2025). Monetary policy transmission in India – A coefficient-restricted VAR approach. *Applied Economics*. <https://doi.org/10.1080/00036846.2025.2551286>
12. Chakravarty, S. L., & Goel, S. (2025). Monetary policy transmission to banks’ interest rates in India: A comparative study of the Base Rate, MCLR and EBLR regimes. *MUDRA: Journal of Finance and Accounting*, 12(2), 70–88. <https://doi.org/10.17492/jpi.mudra.v12i2.1222504>
13. Dubey, Y., & Shahzeb, S. M. (2025). Deciphering India’s monetary policy transmission: A structural VAR perspective. *European Economic Letters*, 15(1), 1885–1903. <https://doi.org/10.52783/ee1.v15i1.2571>
14. Christiano, L. J., Eichenbaum, M., & Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1), 1–45. <https://doi.org/10.1086/426038>
15. Sims, C. A. (1992). Interpreting the macroeconomic time series facts: The effects of monetary policy. *European Economic Review*, 36(5), 975–1000. [https://doi.org/10.1016/0014-2921\(92\)90041-T](https://doi.org/10.1016/0014-2921(92)90041-T)
16. Bernanke, B. S., & Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic Perspectives*, 9(4), 27–48. <https://doi.org/10.1257/jep.9.4.27>
17. Taylor, J. B. (1993). Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39, 195–214. [https://doi.org/10.1016/0167-2231\(93\)90009-L](https://doi.org/10.1016/0167-2231(93)90009-L)
18. International Monetary Fund. (2024). The great tightening: Insights from monetary policy transmission in advanced and emerging market economies. In *World Economic Outlook, October 2024*. International Monetary Fund.

A Data Sources and Replication Materials

All raw source files, the assembled master table (`msi_master_table.csv`), and the Python notebook that reproduces every estimate and figure in this paper (`paper_pipeline.ipynb`) are available in the public repository:

https://github.com/priyanshrm/navigating_the_dilemma

Table 1 documents the origin of each of the 14 input files used by the notebook. “ACE KP” denotes the ACE Knowledge Portal (<https://www.acekp.in>); menu paths are given as they appear in the portal’s left-hand navigation. Coverage dates are taken from each file’s own title line.

Table 1: Data sources for all input files used in the empirical pipeline.

#	File	Role in pipeline	Source and access path	Coverage	Unit and transformation
1	policy_rates.xls	Repo rate (variable 1)	ACE KP → Money and Banking → Policy Rates	Daily, 1995–2026	% p.a.; last value at each quarter-end
2	consumer_price_index.xls	CPI inflation (variable 2)	ACE KP → CPI, WPI & Inflation → Consumer Price index	Monthly, Jan 2013–Dec 2025	General Index (All Groups), Combined; YoY %, averaged over the three months of each quarter
3	components_of_money_stock__monthly_.xls	Broad money M3 (variable 3; MSI and GovShare denominator)	ACE KP → Money Stock → Components of Money Stock (Monthly)	Monthly, 2008–2027	Rs crore; “Broad Money” at quarter-end; log-difference $\times 100$; $\Delta M3$ summed over four quarters for MSI
4	lending___deposit_rates.xls	Bank prime lending rate (variable 4)	ACE KP → Money and Banking → Lending / Deposit rates	Daily, 1994–2026	PLR range midpoint at quarter-end (no MCLR series in file)
5	fpi_data_2000_2025_FINAL.csv	FPI debt inflow (MSI numerator)	NSDL FPI Monitor archive (https://www.fpi.nsdl.co.in/web/Reports/Archive.aspx); daily reports scraped month-by-month with Selenium and merged into a single table	Daily, trading days	Rs crore; “Debt” and “Debt-VRR” net investment columns summed, then summed within each quarter
6	nri_deposits_net_of_inflow_and_outflow.xls	NRI deposit inflow (MSI numerator)	ACE KP → Trade and Balance of Payment → NRI Deposits → NRI Deposits Net of Inflow and Outflow	Monthly, Apr 2009–Feb 2026	USD mn “Total” summed per quarter, converted to Rs crore at the quarterly average USD/INR rate
7	market_borrowings_of_central_and_state_governments.xls	GovShare numerator	ACE KP → Public Finance → Centre and State → Market Borrowings of Central and State Governments	Annual, FY1980-81 to FY2025-26	Rs crore; net borrowing divided by four per quarter, summed over four quarters, divided by $\Delta M3$
8	forex_rate_usd.xls	Exchange rate; USD-to-INR conversion	ACE KP → Forex → Forex Rate (USD)	Daily, 1994–2026	Quarterly mean of the USD column
9	quarterly_gdp_at_current_prices__base_year_201112.xls	Master quarter index; GDP denominator for fiscal deficit and trade	ACE KP → National Income Statistics → Components of GDP → Quarterly GDP at current prices – Base year (2011-12)	Quarterly, Jun 2011–Sep 2025	Rs crore; row “Total Expenditure on GDP”
10	key_deficit_indicators_of_the_central_government.xls	Fiscal deficit, % of GDP (control in master table; not in final models)	ACE KP → Public Finance → Central Government → Key Deficit Indicators of the Central Government	Annual, FY1970-71 to FY2025-26	Gross Fiscal Deficit divided by annual GDP $\times 100$
11	balance_of_payment_in_usd_qrtrly.xls	Current account balance (dropped during cleaning)	ACE KP → Trade and Balance of Payment → Balance of Payment → Balance of Payment in USD (Qrtrly)	Quarterly; data ends 2019Q3	USD mn; removed because of missing observations after 2019Q3

Continued on next page

Table 1 (continued)

#	File	Role in pipeline	Source and access path	Coverage	Unit and transformation
12	indias_foreign_trade_usd_monthly.xls	Trade openness (control in master table; not in final models)	ACE KP → Trade and Balance of Payment → Foreign Trade → India's Foreign Trade USD Monthly	Monthly, 2008–2026	Exports plus imports (USD mn), converted to Rs crore, divided by GDP
13	master_wpi_index.xlsx	WPI food inflation (built in master table; not in final models)	ACE KP → CPI, WPI & Inflation → Wholesale Price index	Monthly	YoY % change in “Food Articles” index, averaged within each quarter
14	brent_crude_fred.csv	Brent crude oil price (control)	Yahoo Finance	Daily	USD per barrel; quarterly average, then YoY % change

Note. The files above are placed in a single folder together with the notebook; the notebook's first cell builds the quarterly master table from them, and subsequent cells run each phase of Section 4.2 in order.